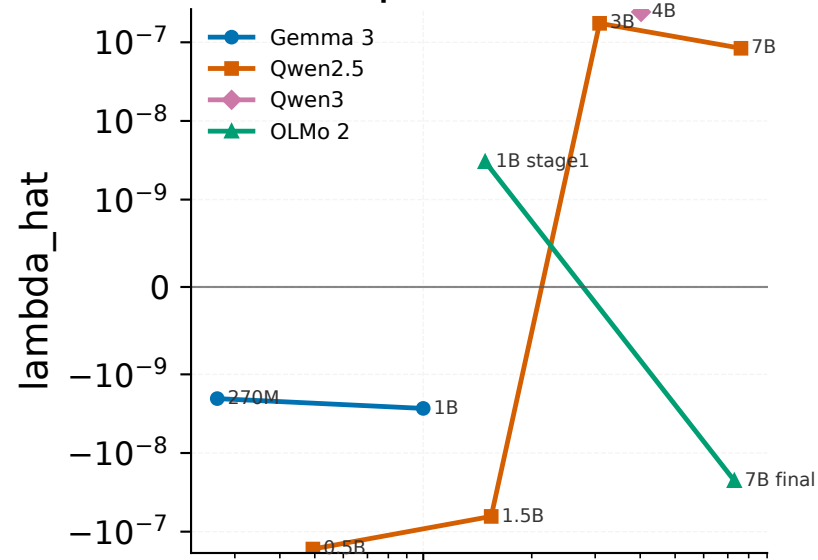
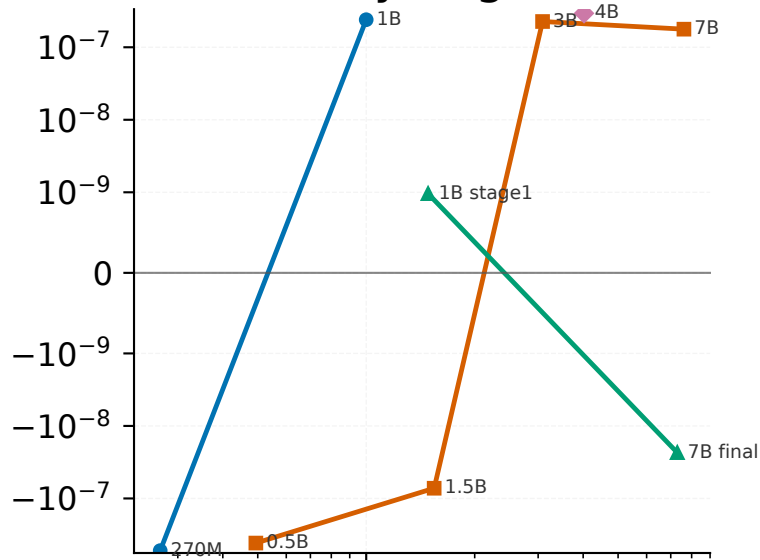


LLM ridge estimates by model size and parameter group

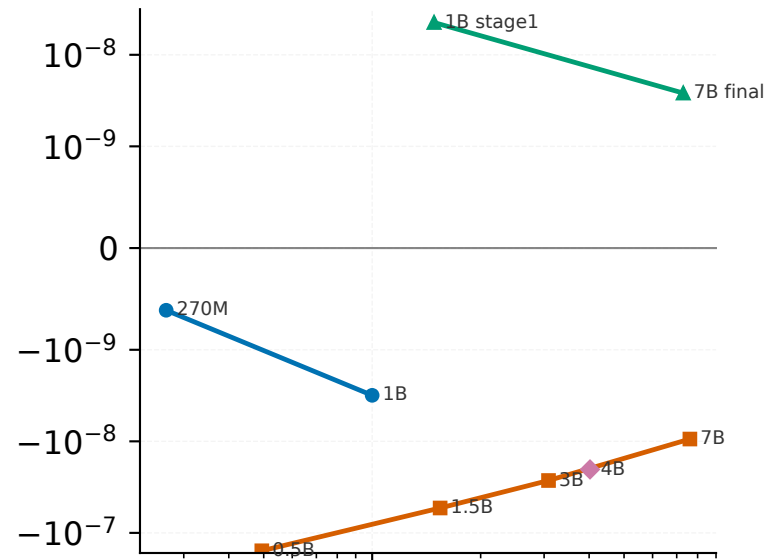
All parameters



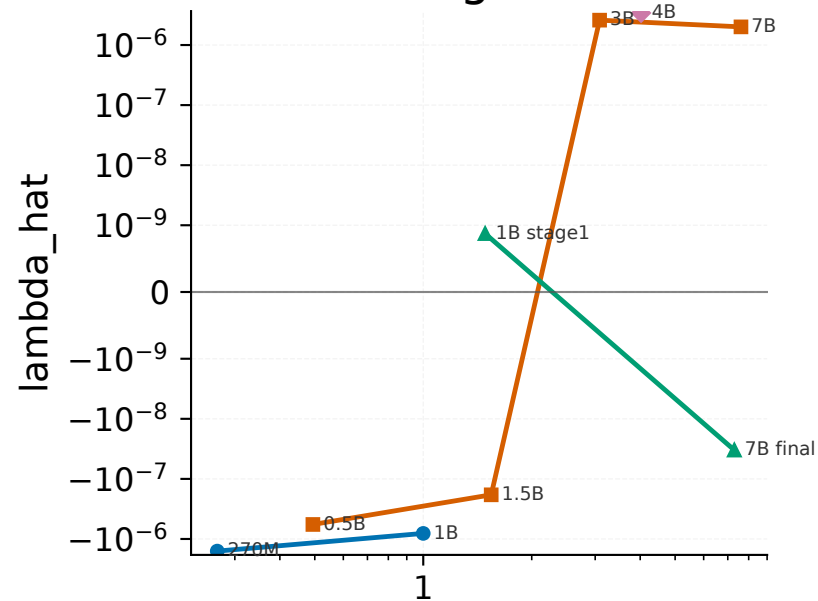
Decay-eligible



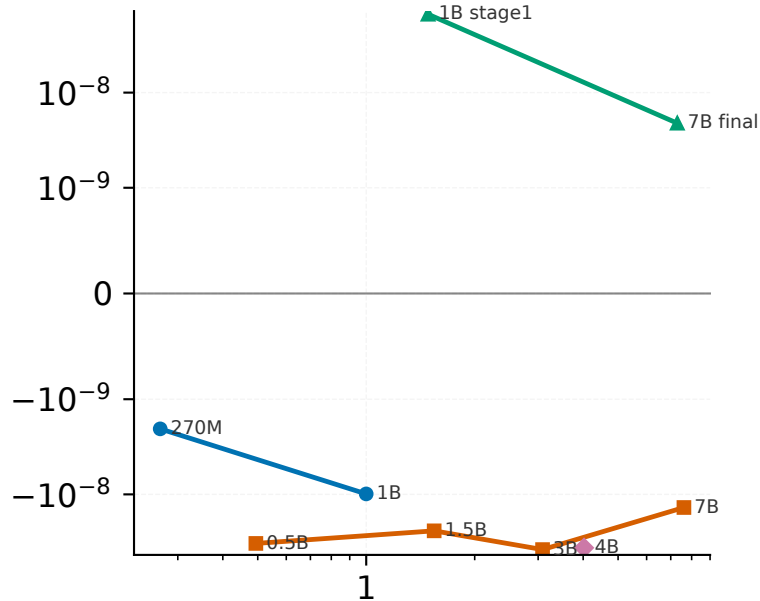
Attention + MLP



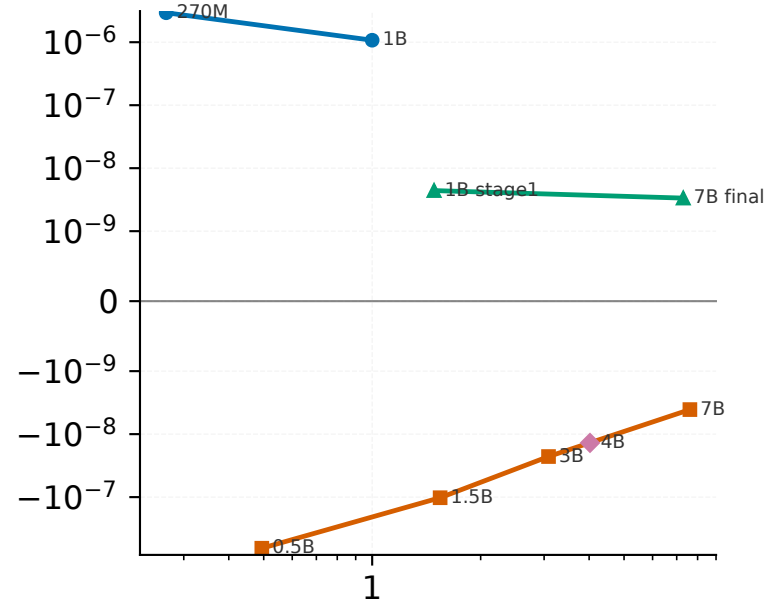
Embedding / LM head



Attention



MLP



Total unique trainable parameters (B)

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X-axis uses signed symlog scaling with a zero reference line; points are 50M-token Wiki shard estimates.